Do Crowding Measures Warn of Factor Losses? Volatility and Business-Cycle Risk*

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Abstract

Crowding reports promise to tell investors which factors are dangerous. Built from returns, they mostly tell them which factors are volatile. We test centrality, comovement, and run-up measures on 104 equity factors in the United States and other developed markets and 148 US anomalies. Centrality ranks factors as volatility does, with a US rank correlation of 0.96. No crowding measure warns of losses beyond volatility; comovement and run-up point to smaller relative losses. Among US factors, a business-cycle beta does flag such losses; outside them, the evidence is weaker. A crowding report built on centrality largely restates the risk report.

Keywords: Factor Investing, Crowding, Centrality, Principal Components, Business Cycle, Risk Budgeting.

JEL codes: G11, G12, E32

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1 Introduction

Ask a crowding report which equity factor theme is the most crowded, and in nearly every month since 1957 it gives the same answer: value. Ask a risk report which theme is the most volatile, and it gives the same answer again. That is no coincidence. Across 104 US equity factors, the most common crowding score ranks factors as volatility does, with a rank correlation of 0.96.

The point matters because many institutions now monitor factor crowding (Chincarini et al. 2026). Positions are observed only for some investors, and crowded trades are hard to identify (Pojarliev and Levich 2011), so crowding is usually read from returns. Three kinds of signal dominate. Centrality measures a factor's share of the portfolio's common risk (Kritzman et al. 2011; Kinlaw et al. 2012, 2019). Comovement measures how closely a factor moves with other factors, or its stocks with each other (Lou and Polk 2022; Baltas 2019). Run-up measures its recent return. Industry models combine such signals with holdings, short interest, and valuation spreads (Bayraktar et al. 2015), and report that crowded factors have often suffered large drawdowns afterward (Bonne and Roisenberg 2019). We test the returns-based ingredients of such reports, not the vendor models themselves.

Yet a volatile factor is not necessarily a crowded one. Centrality is computed from the covariance matrix, and its originators use volatility as a proxy for vulnerability (Kinlaw et al. 2012, endnote 2). Some link to volatility is therefore built in. Capital, however, tends to leave factors after their volatility rises (Hanson and Sunderam 2014; Barroso et al. 2022), so high volatility need not mean crowded positions. The test that matters is whether a crowding measure tells an investor anything that volatility does not. We measure a loss beyond normal risk as a factor's maximum drawdown over the next 12 months divided by its trailing volatility. A measure earns credit only for losses that volatility does not already predict. We then ask what the answers mean for sizing and timing factor sleeves.

We test the measures on 104 equity factors of Jensen et al. (2023) (JKP) and on 148 anomaly portfolios of Chen and Zimmermann (2022), whose daily decile returns also allow comovement within a factor to be measured. The United States comes first: it offers the longest history, ten recessions since 1957, and our business-cycle indicator. Developed markets outside the United States, available from 1990, provide independent stocks but only three recessions. Crowding measures say how much factors move together but not why. We add a direction, the business cycle, in two forms: each factor's cycle beta and, within the crowding framework, target principal component analysis (Bryzgalova et al. 2026), which directs centrality at the cycle. The difference from standard centrality is a factor's *excess centrality*.

Three findings stand out. First, centrality across factors is a volatility ranking in disguise, in every decade, window, and universe: its rank correlation with trailing volatility is 0.96 in the United States, 0.94 in developed markets, and 0.88 on the anomalies. The factors it labels crowded lose in recessions what volatile factors lose (Exhibits 1, 2, and 4).

Second, no returns-based crowding measure warns of losses beyond volatility (Exhibit 3). Centrality adds nothing. Comovement across factors, comovement in down markets, comovement within factors, and past run-ups even point the wrong way: factors that score high on them later lose less, not more, relative to their risk.

Third, among US factors, exposure to the business cycle does warn of such losses. US factors with a large cycle beta later lost more than their volatility implied (rank-regression t of 5.6, and 4.4 with one observation a year), also after controls for centrality, short-term volatility, and run-up. Over the following year, the top 30% of US factors by cycle beta lost 3.2 percentage points more than the bottom 30%, or 0.08 more per unit of annual volatility. Sorts on centrality or volatility show no difference per unit of volatility. Outside the US factors, the evidence is weaker. In developed markets, the cycle beta is not significant (t of 1.7), and the US magnitude is rejected. Among the anomalies, the signal is clearest in

a share-of-risk version of the cycle beta and comes mostly from rising volatility. Excess centrality reproduces the US signal but adds nothing beyond the cycle beta.

For portfolio construction, no crowding measure is a profitable return signal, and crowding caps add little once weights account for risk. Risk-based weights raise the US Sharpe ratio, gross of costs, but not significantly elsewhere. In US recessions, they also lost more per unit of volatility than equal weights.

The lesson for investors fits in three rules. Read a crowding report next to the risk report: if it ranks sleeves by volatility, it adds nothing new. Size sleeves in the risk budget, not with crowding caps. And watch each sleeve's cycle beta next to its volatility: among US factors, it is the one measure here that flags losses beyond normal risk. Crowding scores measure how much risk a factor shares; in US factors, the losses that volatility misses depend on what that risk is tied to.

The article adds to three strands of work. It tests returns-based crowding measures side by side against volatility, in the spirit of evidence in this journal that a widely watched indicator need not carry the risk attributed to it (Kritzman and Turkington 2026). For positioning research (Hanson and Sunderam 2014; Brown et al. 2022; Barroso et al. 2022; Volpati et al. 2020; Chincarini et al. 2026), it shows that returns-based scores cannot be validated as positioning measures from returns alone. And it applies target principal component analysis to risk monitoring. As with risk-premium principal component analysis (Lettau and Pelger 2020), the change to the method is small: the business cycle is the direction that matters for losses, but a simple cycle beta captures it. The tests were developed on the anomalies and recorded in a dated research log before they were computed on the JKP factors; Section 2 states the exceptions.

2 Data and Measures

2.1 Factors in the United States and Other Developed Markets

The main data are the factor returns of Jensen et al. (2023). This public database rebuilds 153 published equity factors consistently for 93 countries. Each factor is a long–short portfolio of the top and bottom terciles of a firm characteristic, signed as in the original study. Breakpoints come from stocks above the 20th NYSE size percentile, and stock weights equal market capitalization capped at the 80th NYSE percentile. These choices make the factors closer to investable strategies than many original studies, and comparable across regions.

We exclude 49 of the 153 factors. The registered design excluded 31 factors whose sorting variable is built from past returns (momentum, reversal, and seasonality), from the price level or size, or from trading activity. It kept 18 factors sorted on statistics of past stock returns, such as betas, volatility, and skewness. After the first results on that 122-factor universe were known, we also removed these 18 factors. The business-cycle target had moved most of the common risk to the low-risk factors among them, which hid how it distributes risk among the other factors. Centrality, itself computed from past returns, is also related to their sorting variables by construction. Section 7 reports the results on 122 and on all 153 factors. Nine JKP clusters, which contain 98 of the 104 factors, serve as themes (Exhibit A1). JKP define them by clustering the factors' US return correlations, so they are a descriptive grouping.

The United States is the main sample, with portfolios formed monthly from January 1957 to December 2024, the end fixed in the research log before the JKP tests. Developed markets outside the United States (22 countries, formation from December 1990) contain independent stocks and serve as the out-of-sample test. They offer only three recessions and about 385 test months, against 660 in the United States, and they are measured against the US cycle. The number of factors grows with data availability, from 73 to 104 in the

United States and from 42 to 104 in developed markets. Emerging markets are reported descriptively.

The second data set is the development sample: the long–short portfolios of 148 anomalies of Chen and Zimmermann (2022) from the Open Source Asset Pricing project, US stocks, formed from January 1957 to December 2024. Unlike the JKP data, the project also publishes the daily returns of the ten decile portfolios behind each anomaly, from which comovement within a factor can be measured. Momentum, reversal, size, liquidity, and volume anomalies are excluded, but 29 anomalies sorted on return seasonality, other statistics of past returns, or the price level are kept (Supplement). The two data sets cover the same US stocks and period, so they are not independent of each other. Exhibit A2 of the Appendix describes all data sets.

We measure the business cycle with an indicator adapted from Swade et al. (2024). It averages five components, each standardized over an expanding window and winsorized at three standard deviations: the price-to-earnings ratio, the negative BAA–AAA credit spread, industrial-production growth, and expected and unexpected inflation. Expected inflation comes from a recursive autoregression of consumer-price changes (Section B of the Appendix). Regimes follow the switch filter of Swade et al.: expansion (indicator positive and improving), peak (positive and deteriorating), recession (negative and deteriorating), and recovery (negative and improving). The indicator uses current data vintages, and industrial-production growth enters as the change into the following month. It is therefore a recursively estimated rather than a strictly real-time measure. Exhibit A3 of the Appendix shows it since 1957; its monthly change is written ΔZ.

2.2 Returns-Based Crowding Measures

The measures fall into three families. The first measures the size of a factor's share of common risk. A principal component decomposition of the trailing returns of N_t factors

gives the centrality of factor i: its share of the leading components, weighted by the variance they explain (Kinlaw et al. 2019),

$$C_i \;=\; \frac{\sum_{j=1}^{K} \sigma_j^2 \cdot |v_i^j| \big/ \sum_{k=1}^{N_t} |v_k^j|}{\sum_{j=1}^{K} \sigma_j^2}, \tag{1}$$

where σ_j^2 is the j-th eigenvalue of the decomposed matrix and v^j its unit eigenvector; for the covariance matrix, these are the variance of component j and the factors' loadings on it. The scores sum to one. Because loadings enter in absolute value, they measure the size of a factor's common exposure, not its direction. Kinlaw et al. (2019) use the score across sectors and within each of four equity factors, and read it as a shift against its own trailing three years. We apply the sector form to the factors themselves, with 60 months of monthly returns and $K = 5$ components, because a multi-factor investor wants to know which sleeves carry the portfolio's common risk. Our conclusions therefore concern the level of centrality across sleeves. Computed from the covariance matrix, centrality mixes how strongly a factor moves with the others and how volatile it is; we call this the *standard score* C^{PCA}. Computed from the correlation matrix, the scale drops out; we call this C^{COR}.

The second family measures comovement directly. The mean correlation of a factor with all other factors over the trailing 60 months is the simplest version. The downside correlation repeats it over months with a negative market return. Comovement within a factor follows the idea that crowded arbitrage makes the stocks on each side of a factor move together beyond common factors (Lou and Polk 2022). Over the trailing 252 trading days, we regress the daily excess returns of the three deciles of the long leg ($\mathcal{H}$: deciles 8 to 10) and of the short leg ($\mathcal{S}$: deciles 1 to 3) on a constant and the daily Fama–French three factors. We then average the pairwise correlations ρ_{jk} of the residuals within each leg,

$$\mathrm{CoMov}_{i,t} \;=\; \tfrac{1}{2}\Big(\overline{\rho}_{jk}^{\,\mathcal{H}} + \overline{\rho}_{jk}^{\,\mathcal{S}}\Big). \tag{2}$$

Three decile portfolios per leg are coarser than the individual stocks of Lou and Polk (2022), an approximation that the public data allow. The third family is past performance: a factor's compounded return over the past 12 months, the trailing-performance input of industry crowding models (Bonne and Roisenberg 2019), which use 36 months. The mean and downside correlations and the run-up are our own simple versions of these ideas.

2.3 Business-Cycle Exposure

Factors can move together because of the business cycle, liquidity or funding shocks, or shared positions. The indicator measures the first, and the simplest measure of a factor's exposure to it is its cycle beta $|\lambda_i|$. This is the absolute slope of a regression of the factor's monthly returns on the innovation of the indicator, the part of its monthly change that a first-order autoregression does not predict. Both regressions use all months up to three months before the formation date, which covers the publication lags of the inputs, and the cycle beta requires at least 36 months. The absolute value measures the size of a factor's cycle exposure, as centrality measures the size of its common exposure; the sign is examined in Section 5. The *cycle share* turns the cycle beta into a share of common risk: each factor's share of the summed $|\lambda|$ across factors, minus its standard centrality. It is therefore less related to volatility than the cycle beta itself.

The methodological step of this article is to bring the cycle into the crowding framework itself. Standard centrality decomposes the covariance matrix of the trailing returns, whatever drives their comovement. Target principal component analysis (Bryzgalova et al. 2026) decomposes a matrix that is tilted toward economic information. Let $R \in \mathbb{R}^{T\times N}$ hold the trailing $T = 60$ months of returns of the N factors, demeaned within the window. The method applies principal component analysis to

$$\Omega \;=\; \frac{1}{NT}\left(I_N + \tilde{\gamma}_{XS}P_H\right) R^{\top}\left(I_T + \gamma_{TS}P_D\right) R \left(I_N + \tilde{\gamma}_{XS}P_H\right), \tag{3}$$

where P_H projects onto a cross-sectional target H and P_D onto a time-series target D. The cross-sectional target is the vector of the factors' cycle loadings λ. It strengthens directions along which factors with similar cycle exposure move together. The time-series target is the series of the indicator's downturns over the previous year, $\min(\Delta_{12}Z, 0)$. It strengthens components whose returns move with these downturns. With both strengths at zero, Ω is proportional to the covariance matrix, and the method returns standard centrality exactly. The strengths are fixed at the values of the development sample, $\gamma_{XS} = 0.1$ and $\gamma_{TS} = 10$, and were not tuned on the JKP data; $\tilde{\gamma}_{XS}$ rescales γ_{XS} for the number of factors (Section B of the Appendix). With one cross-sectional target and 104 factors, these strengths multiply the variance along the direction of the cycle loadings by about 11, and by about 16 with 148 anomalies. The targeted decomposition is then close to a share of the cycle loadings, the time-series target has little effect, and excess centrality should behave much like the cycle share. Centrality computed from Ω with Equation (1) is the macro-targeted centrality $C^{\text{XS-TS},c}$. It has the same scale as standard centrality, so the two can be compared factor by factor. The excess centrality of factor i is

$$\Delta C_{i,t} \;=\; C_{i,t}^{\text{XS-TS},c} - C_{i,t}^{\text{PCA}}. \tag{4}$$

It sums to zero across factors. A positive value marks a sleeve whose common risk is more tied to the cycle than its standard share suggests.

2.4 Outcomes and Tests

The main outcome is a factor's loss beyond normal risk: its maximum drawdown over the next 12 months, compounded from monthly returns and recorded as a positive number, divided by its trailing 60-month volatility. A second outcome is the factor's future exposure to the cycle. For each factor and month, we regress its returns over the next 24 months on

ΔZ,

$$r_{i,s} \;=\; a_{i,t} + b_{i,t}\,\Delta Z_s + \varepsilon_{i,s}, \qquad s = t+1, \ldots, t+24, \tag{5}$$

and record its future cycle sensitivity $|b_{i,t}|$, written $|\beta(\Delta Z)|$, and the cycle's share of its variance, $R^2(\Delta Z)$.

Each month, we relate the rank of each measure to the rank of the outcome across factors, alone and with the ranks of measures an investor already has. These controls are trailing volatility, the absolute beta on the regional equity market, the factor's past cycle sensitivity over the same window without its last three months, and the size of its loadings on the five indicator components. The coefficient on the measure is the information it adds. A coefficient of 0.1 means that moving a factor from the bottom to the top of the measure's ranking raises the expected rank of its outcome by about ten percentiles. US tests start in 1968, a date fixed in advance, and end with formation in December 2022, the last month with a complete 24-month outcome window; the loss tests use the same end. Monthly coefficients are averaged with Newey–West standard errors that account for the overlapping windows (Fama and MacBeth 1973; Newey and West 1987): 12 lags for losses and 24 for cycle exposure. The overlap leaves only about 16 to 55 independent observations per region, which limits the power of every test. For the main business-cycle results we also report t-statistics from one observation a year, averaged over the twelve possible starting months. The decision rule for the crowding measures corrects for testing five measures (Holm 1979); the business-cycle measures were added later and belong to no corrected family. For recession sorts, we report in how many of the recessions the gap is negative.

Every test on the JKP data was recorded in a dated research log before it was computed, with the following exceptions, and every test is reported whatever its outcome. The JKP data became the main data set after a first test carried over from the development sample had failed on them. The universe was restricted from 122 to 104 factors after the results on the wider universe were known. The comparison of crowding measures for losses was added

last, and the article was then organized around it; its measures, outcome, and decision rule were recorded before it was computed. The run-up, the downside correlation, the cycle beta, the cycle share, and the long-window cycle beta used as a control followed its first results. After the main results came the comparisons with the cycle share and with more flexible controls, and the checks requested by independent reviewers: the anomalies without return-based sorts, the shrunk covariance matrix for equal risk contribution, the lagged regimes, and the checks of the cycle beta in Sections 5 and 7. All later additions were recorded before they were computed, some with values already seen in exploratory runs, and are labelled as such.

2.5 Portfolios and Backtests

The backtests use three kinds of portfolio. All weights are set at the end of month $t-1$ from the trailing 60 months and held over month t. Returns are gross of trading costs; turnover is one-way, after the drift of last month's weights.

The first kind are sorts. In the month before each NBER recession and at the peak of each market sell-off, the factors are sorted on a measure into the top and bottom 30%. Each group is compared by the average within-episode maximum drawdown of its factors, their absolute cumulative return, and the mean pairwise correlation of their monthly returns during the episode.

The second kind are crowding tilts, which test whether a measure is a return signal. The equal-weighted factor portfolio is tilted toward factors whose score is high relative to its own recent history. Each factor's score is ranked within its trailing 12 months; these percentiles are ranked across factors and centered, $z_{i,t} \in [-\frac{1}{2}, \frac{1}{2}]$; and

$$w_{i,t} \;\propto\; 1 + 0.5\, z_{i,t} \;\text{ for the top and bottom 30\%}, \qquad w_{i,t} \propto 1 \text{ otherwise}, \tag{6}$$

with weights renormalized to sum to one, so that the tilts remain long-only (see Kelly et al. 2019, for the rank transformation). A tilt is judged by its active return over the equal-weighted portfolio and by its alpha after its exposure to that portfolio. Because the score is measured relative to its own history, this test captures changes in crowding, and it complements the comparisons of levels across factors.

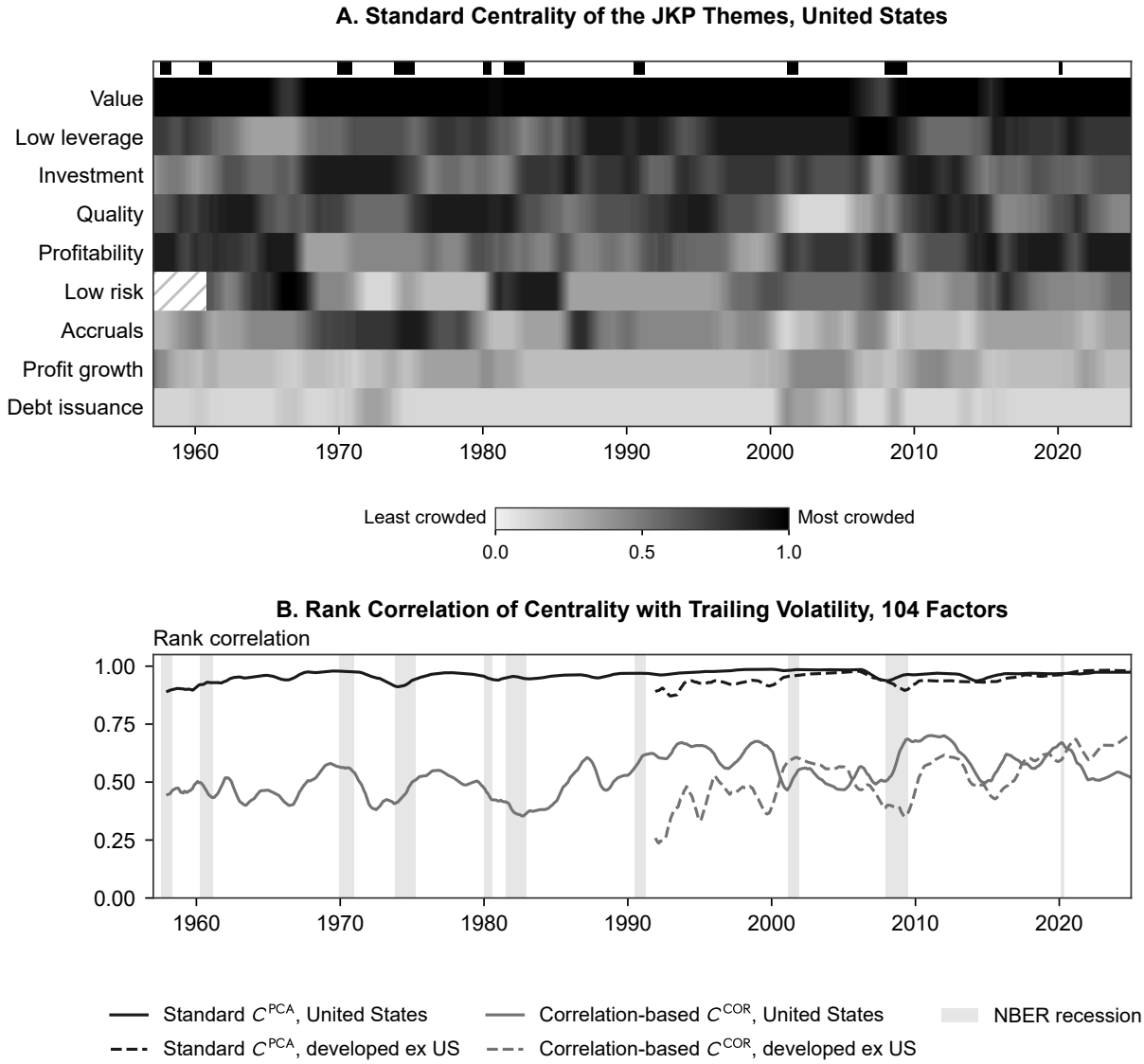
The third kind are sizing rules. Equal weights, inverse-volatility weights (proportional to one over each factor's trailing volatility), and long-only equal risk contribution (Maillard et al. 2010) are each run with and without a crowding cap. Equal risk contribution uses the covariance matrix of the trailing 60 months shrunk as in Ledoit and Wolf (2004), because with up to 104 factors and 60 months the sample covariance matrix is singular. The cap halves the weight of the 30% of factors with the highest C^{PCA} or C^{COR}, after which the weights are renormalized. Differences in Sharpe ratios are tested with the robust test of Ledoit and Wolf (2008), Holm-adjusted where several caps are compared on the same rule. Because losses in downturns are the question of the article, each rule is also judged by its mean maximum drawdown during NBER recessions divided by its annualized volatility.

3 Crowding Scores Rank Factors by Volatility

A crowding report based on standard centrality shows a strikingly stable map (Exhibit 1, Panel A). Value is the most crowded theme in nearly every month since 1957; debt issuance and profit growth are the least crowded. Value is also the most volatile theme, at 10.8% a year against 2.4% for debt issuance (Exhibit A1). Across the 104 factors, the rank correlation between standard centrality and trailing volatility stays between about 0.87 and 0.99 in every decade and in both regions (Panel B).

Exhibit 2 extends the comparison to every measure. Standard centrality is a volatility ranking in all three samples (0.88 to 0.96), and correlation-based centrality removes part of

Exhibit 1. The Crowding Map and Its Volatility Content

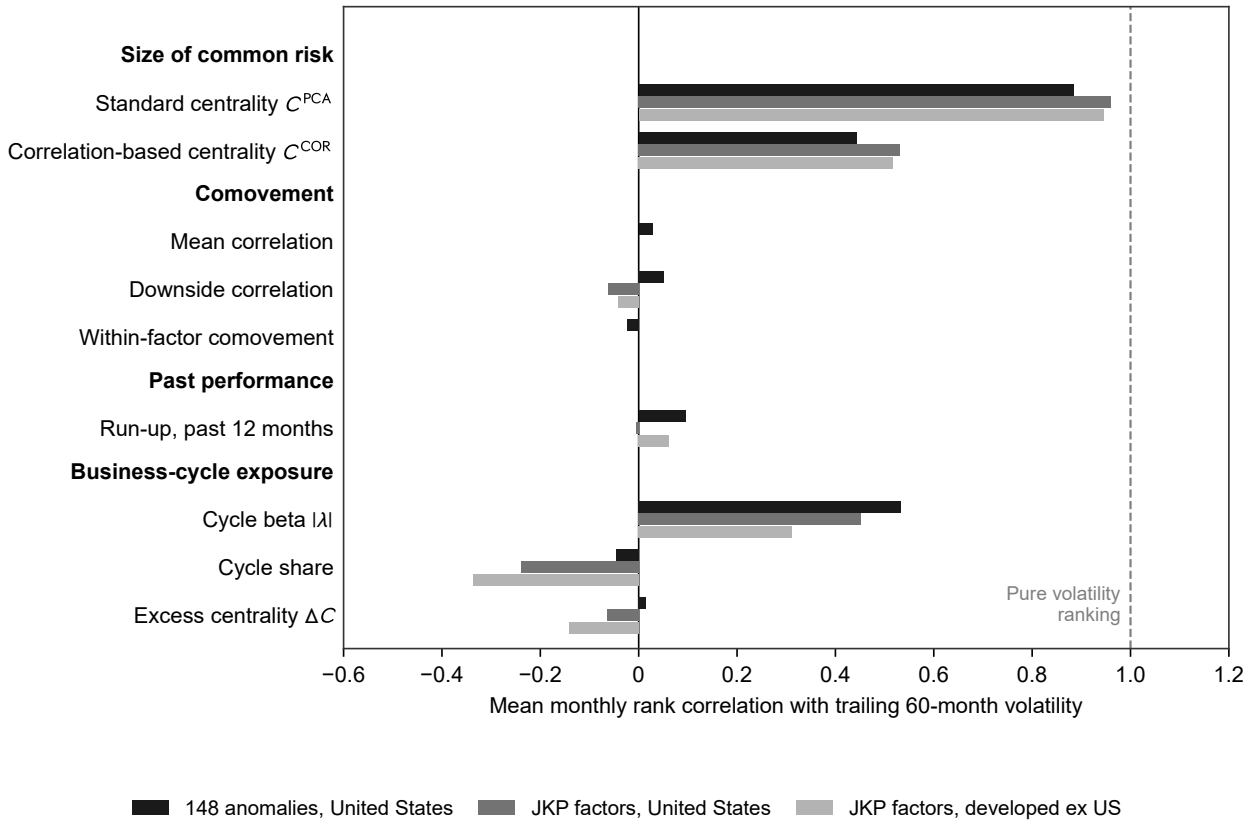


Notes: Panel A: cross-sectional percentile rank of standard centrality C^{PCA} of the nine JKP themes (United States), 12-month trailing mean; hatched: fewer than 60 months of returns; black bars above the map: NBER recessions. Panel B: cross-sectional rank correlation of C^{PCA} and of C^{COR} with trailing 60-month volatility across the 104 JKP factors, 12-month trailing mean; shaded: NBER recessions.

the scale (0.44 to 0.53). The comovement and run-up measures and excess centrality are nearly unrelated to volatility. The cycle beta sits in between (0.31 to 0.53), because volatile factors load more on the cycle in absolute terms. The cycle share is negatively related to volatility on the JKP factors, mechanically, because it subtracts standard centrality.

Exhibit A19 of the Appendix asks what the centrality scores add to measures an investor

Exhibit 2. Volatility Content of the Crowding and Business-Cycle Measures



Notes: Mean monthly cross-sectional rank correlation of each measure with trailing 60-month volatility; a value of one means that the measure ranks factors as volatility does. Anomalies: formation 1968–2024; JKP factors: formation from 1957 (United States) and 1990 (developed ex US). Within-factor comovement requires the daily decile returns of the anomalies; mean correlation was computed for the anomalies only.

already has. Standard centrality is related to a factor's future cycle sensitivity (rank correlation of 0.30), as a volatility ranking must be: volatile factors have larger cycle sensitivities in absolute terms. Beyond volatility, market beta, past cycle sensitivity, and the macroeconomic loadings, however, it adds nothing, neither for cycle sensitivity, where its US coefficient is even negative, nor for the cycle's share of variance (Panels B and C). Correlation-based centrality adds nothing either. Volatility targeting turns the standard score into correlation-based centrality, so this result also covers volatility-targeted sleeves. Read as a shift against its own history, as Kinlaw et al. (2019) use it, the score still tracks volatility: 12-month changes in centrality have a rank correlation of 0.69 with 12-month changes in volatility in the United States and 0.70 in developed markets (Exhibit A17).

The score ends up measuring volatility for two reasons. Its construction lets volatility in on purpose, and volatility differs more across factor portfolios than their correlations do, so the score sorts factors by volatility before it sorts them by how much of their risk is common. Its authors read high volatility as the footprint of crowded trading (Kinlaw et al. 2019). The positioning evidence is mixed: crowded positions are associated with tail risk in market distress (Brown et al. 2022; Chincarini et al. 2026), but capital leaves value strategies, and institutions leave momentum, after volatility rises (Hanson and Sunderam 2014; Barroso et al. 2022). A score dominated by volatility cannot tell a factor that is volatile because many investors trade it from a factor that is volatile because its firms are risky.

4 No Returns-Based Crowding Measure Warns of Losses beyond Volatility

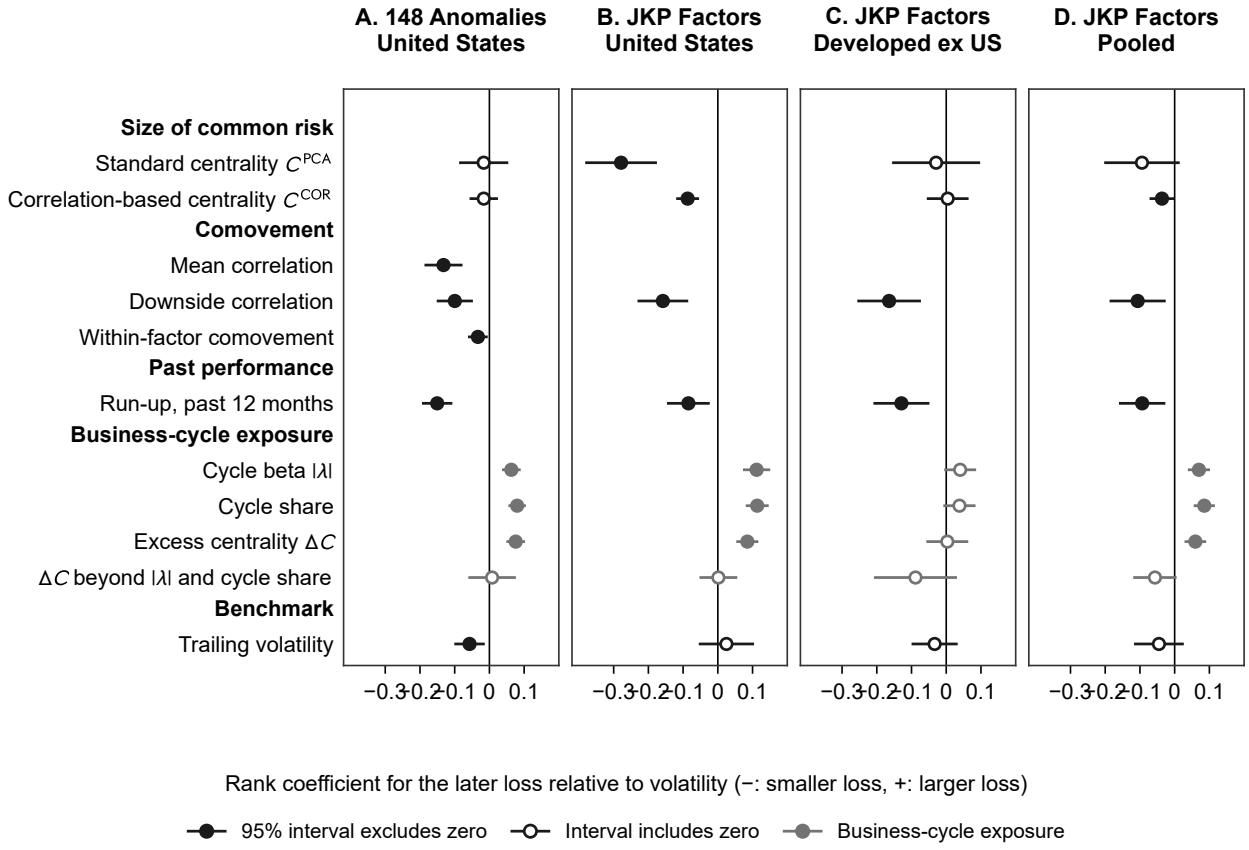
A crowding measure that captured something beyond volatility should flag the factors that later lose more than their normal risk implies. We test this in monthly rank regressions, in the ten NBER recessions since 1957, and in market sell-offs outside recessions.

4.1 Comparison of the Measures

Exhibit 3 tests the measures side by side on the anomalies and on the JKP factors in both regions; Exhibit A20 of the Appendix gives the coefficients. Each coefficient is the information a measure adds about the later loss relative to volatility, beyond the controls of Section 2.

The size measures add nothing. On the anomalies, standard and correlation-based centrality both have coefficients of -0.016. For the raw drawdown, which is not scaled by volatility, volatility itself carries the information, and centrality adds nothing beyond it.

Exhibit 3. Crowding Measures and Later Losses beyond Volatility



Notes: Incremental rank-regression coefficient of each measure for the factor's maximum drawdown over the next 12 months divided by its trailing 60-month volatility, with controls for trailing volatility, absolute market beta, trailing cycle sensitivity, and the macroeconomic loadings; 95% intervals from Newey–West standard errors (12 lags). Positive: larger later losses relative to volatility. Formation 1968–2022 (anomalies, JKP United States) and 1990–2022 (developed ex US, pooled). Filled: interval excludes zero; gray: business-cycle exposure. Values: Exhibit A20.

This is why raw drawdowns of crowded factors can look large without telling an investor anything the risk model does not. On the JKP factors, standard centrality points to smaller relative losses in the United States (t of -5.3) and is insignificant in developed markets. Because centrality and volatility are almost the same ranking, the US coefficient reflects the correlation part of centrality, given volatility, which also enters correlation-based centrality with a negative sign.

The comovement and past-performance measures point the other way. On the anomalies, factors that moved more with the other factors (t of -4.7) or with them in down markets (t of -3.8) later lost less relative to their volatility. So, weakly, did factors whose stocks moved

more together within each leg, although not significantly after the correction for testing five measures. Factors with a strong run-up over the past 12 months also lost less (t of -6.7). On the JKP factors, the downside correlation and the run-up point the same way in both regions and pooled, and on the anomalies the signs are the same after 1990 (Exhibit A20 of the Appendix). The decision rule recorded for this comparison required a positive coefficient with a t-statistic of at least 2 that survives the correction for multiple testing. None of the returns-based crowding measures meets it.

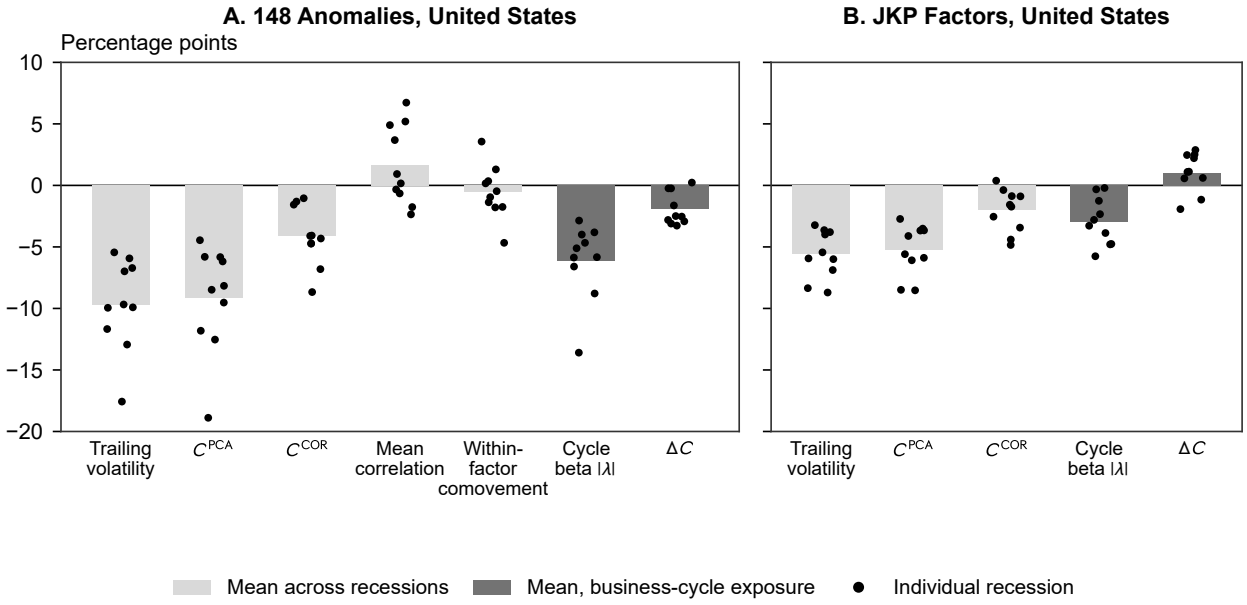
4.2 Possible Mechanisms behind the Smaller Losses

The wrong-way sign is less surprising than it looks. Baltas (2019) reports that, after crowded periods measured from comovement, divergence premia such as momentum tend to underperform, whereas convergence premia such as value, which dominate the factors here, tend to outperform. Two further mechanisms fit the evidence, although the data do not test them. Factors dominated by common movements may have fewer idiosyncratic crashes. And because the outcome divides by a slow 60-month volatility, factors that have just run up, or whose risk has just been low, can look safer relative to it. The run-up result also fits factor momentum (Gupta and Kelly 2019; Ehsani and Linnainmaa 2022). Splitting the outcome helps to tell these mechanisms apart. The loss relative to trailing volatility is the drawdown per unit of next-year realized volatility times the ratio of that volatility to trailing volatility. The comovement and run-up measures predict the first part, not the second. On the anomalies, for example, a high run-up predicts a smaller drawdown per unit of realized volatility (-0.18, t of -7.2) and no change in volatility (t of -0.2; Exhibit A14 of the Appendix). Factors that score high on these measures thus lose less per unit of the risk they actually take, which fits drift and momentum better than a stale denominator. Whatever the mechanism, comovement and run-up readings that a crowding report flags as dangerous have, in these data, preceded smaller rather than larger losses relative to risk.

4.3 Recessions

Recessions are the episodes in which a factor portfolio's common exposure matters most. Exhibit 4 sorts the factors on each measure in the month before each NBER recession and compares the drawdowns of the top and bottom 30% through it. Because recession dates are known only after the fact, the sorts describe which factors were crowded when each recession began, not a tradable signal.

Exhibit 4. Top-Ranked Factors in NBER Recessions



Notes: Maximum drawdown during each NBER recession of the 30% of factors with the highest value of each measure in the month before the recession, minus that of the 30% with the lowest, in percentage points; dots: the ten recessions since 1957; columns: their mean. Negative: the top group fell more. Gray: business-cycle measures. Panel A: 148 anomalies; Panel B: 104 JKP factors, United States. Details: Exhibit A21.

The most crowded JKP factors by standard centrality fell more than the least crowded in all ten US recessions (−9.2% against −4.0% on average). A volatility sort gives the same result, recession by recession: a gap of 5.6 points against 5.2 for the crowding sort (Exhibit A8). On the anomalies, the two sorts are again alike (−9.7 and −9.2 points), and correlation-based centrality halves the gap. The crowded factors did not even move together: their pairwise correlation during recessions was 0.04, against 0.14 for the least crowded (t of −2.0 for the difference). Centrality uses the absolute size of the loadings, so factors

on opposite sides of the same common component are both central. Neither comovement measure picks factors that fell more.

4.4 Market Sell-Offs outside Recessions

If standard centrality measured crowded positions rather than volatility, the difference should show in market sell-offs outside recessions, when crowded investors exit together. Nine US market drawdowns of at least 15% since 1957 do not overlap an NBER recession. They include 1987, the 1998 Russian and LTCM crisis, the 2011 euro-area crisis, and 2022; the selection rule was recorded before the analysis (Exhibit A9). In these episodes, the most crowded factors fell 2.7 percentage points more than the least crowded on average (t of -3.7). A volatility sort gives a slightly larger gap (-2.9 points), and the crowding gap is the smaller one in six of the nine episodes. Neither sort identifies factors that moved together. Over the year after the market peak, the most crowded group earned 2.4 log-return points more than the least crowded (Exhibit A10 of the Appendix), so cutting crowded sleeves at the peak would not have helped on average.

5 Business-Cycle Exposure Signals Losses beyond Normal Risk in US Factors

Crowding measures describe how much a factor moves with the others. Business-cycle measures describe what it moves with. Among US factors, that difference matters for losses.

5.1 The Cycle Beta

The gray markers in Exhibit 3 show the business-cycle measures. Among the US JKP factors, the cycle beta predicts losses beyond normal risk (0.112, t of 5.6). Moving a factor from the

bottom to the top of the cycle-beta ranking raises the expected rank of its later loss, relative to its volatility, by about 11 percentiles. The result does not rest on overlapping windows: with one observation a year, the t-statistic is 4.4 on average across starting months and at least 4.0 for each. It survives added controls for standard centrality, 12-month volatility, and the past run-up. It is not a theme effect, because it holds relative to the theme mean. Nor is it publication decay: it holds among factor-months after each factor's publication and with the years since publication as a control. In investor units, the 30% of US factors with the highest cycle beta lost 3.2 percentage points more over the next year than the 30% with the lowest, or 0.08 more per unit of annual volatility. Sorts on centrality or on volatility give larger raw differences (5.2 and 5.3 points) but none per unit of volatility. These checks were added after the main results; Exhibit A14 of the Appendix reports them.

Splitting the outcome shows what the signal predicts. Among US factors, a high cycle beta predicts both a larger drawdown per unit of the volatility realized over the next year (0.089, t of 4.5) and a rise in volatility above its five-year level (0.062, t of 3.3). It is present in 12-month windows with and without a recession (Exhibit A14). The recession sorts tell a more modest story. The US factors with the highest cycle beta fell more than those with the lowest in all ten recessions, by 2.9 percentage points on average (Exhibit 4; sign test p of 0.002). But this is mostly the volatility content of the cycle beta. Ranked on the part of the cycle beta unrelated to volatility, the top group fell only 0.8 points more, and in seven of ten recessions. Nor does the sign of the loading help: the most procyclical factors did not fall more in recessions than the most countercyclical ones (a gap of +0.9 points). The cycle beta is therefore a flag for the size of a factor's cycle exposure, not a forecast of which factors fall in the next recession.

The anomalies confirm the signal only in part. On the 148 anomalies, the cycle beta predicts losses beyond normal risk (0.063, t of 4.6), and the cycle share does so more strongly (0.080, t of 6.2). The anomalies with the highest cycle beta fell more in all ten recessions, also when the cycle beta is purged of volatility (by 2.7 points). Two qualifications apply. The

anomaly signal comes mainly from windows without a recession and from rising volatility rather than larger drawdowns per unit of realized risk. And without the 29 anomalies sorted on return seasonality, other statistics of past returns, or the price level, the cycle beta weakens to 0.027 (t of 1.8). The cycle share (0.056, t of 4.2) and excess centrality (0.054, t of 3.5) keep the signal (Section 7).

For US factors, the economic reading is direct: exposure to the business cycle, in either direction, rather than the size of a factor's common risk, shows up as losses beyond normal risk.

5.2 From Centrality to Cycle-Directed Centrality

A crowding report built on centrality cannot see this exposure. The leading components of a factor portfolio mix the business cycle with other sources of comovement, and centrality measures the size of a factor's loadings on all of them. Target principal component analysis (Equation (3)) gives the decomposition a direction while keeping the centrality formula and its scale. The design used here is the second one the project tested. The first, carried over from the development sample, applied the targets to undemeaned returns; on the JKP factors it barely moved the crowding map, and its active returns came mostly from average returns (Section E of the Appendix). The second design demeans the returns and estimates the cycle loadings over all available history. Only its cross-sectional target carries information (Exhibit A16, Panel B).

5.3 Excess Centrality Adds Little beyond the Cycle Beta

Excess centrality reproduces the US loss signal: 0.076 (t of 5.5) on the anomalies and 0.085 (t of 5.2) on the JKP factors. In developed markets it is zero. Its signal is not centrality with the sign reversed: adding standard centrality as a control leaves it unchanged (0.077, t of

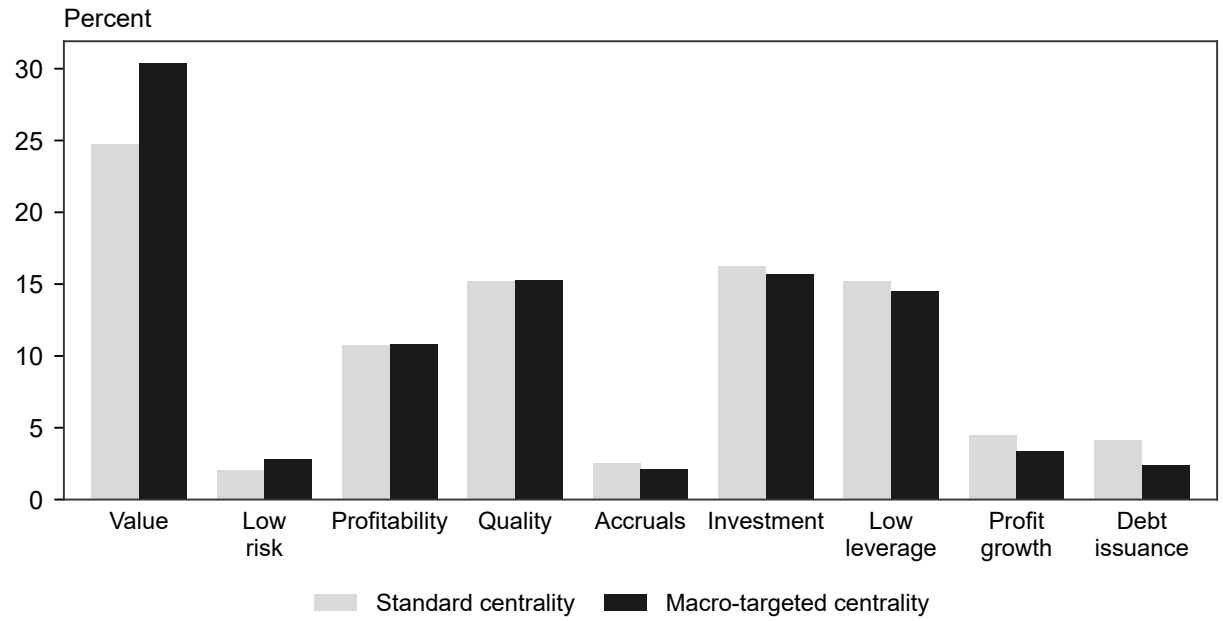
4.5), and the same holds for the cycle share. Beyond the cycle beta alone, excess centrality retains some information in the United States, with t-statistics of 3.8 on the anomalies and 2.9 on the JKP factors, but not pooled. Beyond the cycle beta and the cycle share together, it adds nothing (t of 0.2 on the anomalies and 0.1 on the JKP factors). Unlike the cycle beta, excess centrality is no recession signal: the JKP factors with the highest excess centrality fell 1.0 point less than the lowest-ranked in US recessions. For future cycle sensitivity, the picture is the same (Exhibit A17 of the Appendix). The two scores also pick largely the same factors: 64% of the US factors in the top 30% by excess centrality are in the top 30% by the cycle beta (71% in developed markets). As the strength of the target implies (Section 2), the practical content of the cycle target for an investor is the cycle beta itself.

What excess centrality does add is a picture of where cyclical exposure sits in the portfolio. Directing the decomposition at the cycle lowers the rank correlation of centrality with volatility to 0.83 in the United States and 0.69 in developed markets (Exhibit A19, Panel A). In November 2007, a month fixed in advance, the target raised the share of value in the portfolio's common risk from 24.8% to 30.4% and lowered the shares of debt issuance, profit growth, low leverage, and investment (Exhibit 5). Over time, quality has the highest excess centrality in most months until 2009, and value in most months after 2010 (Exhibit A22 of the Appendix). Because excess centrality sums to zero, the map also depends on the universe (Sections 2.1 and 7). It describes where cyclical comovement sits within a given set of sleeves; it is not a statement about the economy.

5.4 The Cycle Target Selects the Same Factors in Every Phase

A rotation strategy would want the cycle target to select different factors in different phases of the cycle. It largely does not. Averaged within each regime of the indicator, the same US factors carry the highest excess centrality in every phase: the safety component of quality minus junk (3.2 to 3.5 percentage points of common risk), net equity payout and dividend

Exhibit 5. Theme Shares of Common Risk under Standard and Cycle-Targeted Centrality, November 2007



Notes: Share of each JKP theme (United States) in the common risk of the 104 factors under standard centrality and under macro-targeted centrality, November 2007 (sum over a theme's factors). Excess centrality is the difference. The nine themes cover 98 of the 104 factors; the six kept factors of the seasonality and size clusters are not shown.

yield from the value theme (1.3 to 1.5 points each), and earnings volatility. Research and development to sales and asset liquidity carry the lowest. What changes with the phase is the size of the shift (Exhibit A23 of the Appendix). The target raises the share of value most in recessions and recoveries (by 2.7 and 2.2 percentage points, against 1.7 in expansions and 1.4 at the peak) and that of quality most at the peak (2.1 points). In developed markets, it lowers the share of value in every regime. The performance of these selections does not depend reliably on the phase either. After the five Fama–French factors and a correction for testing nine themes, no theme's performance differs significantly across regimes (Exhibit A6 of the Appendix). The tilt toward excess centrality earned most in US recoveries (0.13% a year, t of 2.3), the effect the development sample had pointed to and that a test recorded in advance was set up to check. But recoveries cover only 25 months, 28% of random shifts of the regime sequence produce a return at least as large, the effect disappears when the regimes are lagged by two months (t of 1.4), and in developed markets the recovery return is negative.

The cycle target describes a stable allocation of cyclical risk, not a phase-dependent selection to rotate on.

5.5 The United States and Other Developed Markets

Outside the United States, the cycle beta is positive but not significant (0.040, t of 1.7, and 1.2 with one observation a year), and excess centrality is zero. This is not only a matter of power. Over the common period 1990–2022, the US coefficient is 0.131 (t of 4.6), and the difference between the regions is significant (0.091, t of 2.3), so the US magnitude is rejected in developed markets. What remains of the signal there is mainly a forecast of rising volatility (0.057, t of 2.9), not of larger drawdowns per unit of realized risk (0.010, t of 0.4). Pooled results (t of 4.3) combine the two regions and add no separate evidence. One reason may be that developed-market factors are measured against the US cycle, which describes their own economies only in part; with the broad FRED-MD macroeconomic target instead, the excess-centrality signal is stronger in developed markets than in the United States (Section 7). Until regional business-cycle indicators are tested, the loss signal of business-cycle exposure is a US result.

6 Risk Budgets, Not Crowding Caps, Size Factor Sleeves

6.1 Crowding as a Return Signal

The crowding tilts of Section 2.5 test whether a score that is high relative to its own recent history is a return signal.

No crowding tilt earns a meaningful active return (Exhibit 6). Tilts toward standard crowding raise the exposure to the equal-weighted factor portfolio and earn less than that

Exhibit 6. Crowding Tilts and Factor Returns

	Active		On equal-weighted		Alpha of active return after		
Tilt toward	return (%)	IR	β	α (%)	Volatility	Persistence	Both
Panel A: US, 1958–2024							
C^{PCA}	0.00	−0.01	1.05	−0.13 (−4.8)	−0.02 (−0.9)	−0.01 (−0.4)	−0.03 (−1.3)
C^{COR}	−0.02	−0.14	0.99	0.01 (0.2)	−0.02 (−1.0)	−0.03 (−1.6)	−0.03 (−1.7)
ΔC	0.07**	0.30	1.01	0.04 (1.1)	0.07 (2.2)	0.04 (1.9)	0.04 (1.9)
Volatility tilt	0.02	0.10	1.03	−0.05 (−2.3)			
Premium persistence tilt	0.29**	0.49	1.00	0.30 (1.7)			
Panel B: Developed ex US, 1991–2024							
C^{PCA}	−0.03	−0.15	1.06	−0.16 (−4.3)	−0.01 (−0.4)	0.00 (−0.1)	0.01 (0.4)
C^{COR}	0.02	0.15	1.00	0.02 (0.8)	0.01 (0.7)	−0.01 (−0.7)	−0.01 (−0.7)
ΔC	0.00	0.00	1.00	0.00 (0.1)	0.00 (0.0)	−0.05 (−1.9)	−0.05 (−1.8)
Volatility tilt	−0.02	−0.14	1.04	−0.10 (−4.8)			
Premium persistence tilt	0.24**	0.66	1.02	0.20 (1.4)			

Notes: JKP factors. Long-only tilts toward each signal (Equation 6) relative to the equal-weighted factor portfolio. Active return: annualized mean difference; * and ** denote significance at the 5% and 1% levels (one-sided block bootstrap). IR: information ratio. β and α: slope and annualized intercept on the equal-weighted portfolio. Alpha after: intercept after controlling for the active returns of tilts toward trailing volatility, premium persistence (trailing 60-month mean), or both. Newey–West t in parentheses (6 lags in the United States and 5 in developed markets, by the rule of Newey and West 1994). Common months of all tilts; the US starts in 1958 because the tilt needs a 12-month score history.

exposure implies, an alpha of −0.13% a year in the United States and −0.16% in developed markets. The effect is precisely estimated but economically small, and it disappears once the active return of a tilt toward trailing volatility is controlled for: the crowding tilt behaves as a volatility tilt. Tilts toward correlation-based centrality earn nothing in either region. A tilt toward excess centrality earns 0.07% a year in the United States and nothing in developed markets, and its US alpha is not significant. Because the tilts move only part of the portfolio, they have little power to detect a return signal. A cruder test, added after the main results, compares the equal-weighted return of the 30% of factors with the highest score with that of the 30% with the lowest. Standard centrality again shows no significant difference. For correlation-based centrality, the result is the opposite of a crowding discount: the most central factors earned 1.6% a year more in the United States and 1.5% more in developed markets. Factors with a high cycle share earned 1.5% a year less in the United States, in line with their larger losses (Exhibit A14). By these measures, crowded factors did

not earn less. Only the persistence of factor premia earns a material active return: 0.29% a year in the United States and 0.24% in developed markets, with information ratios of 0.49 and 0.66. This is consistent with, though slower than, the factor momentum documented by Gupta and Kelly (2019) and Ehsani and Linnainmaa (2022).

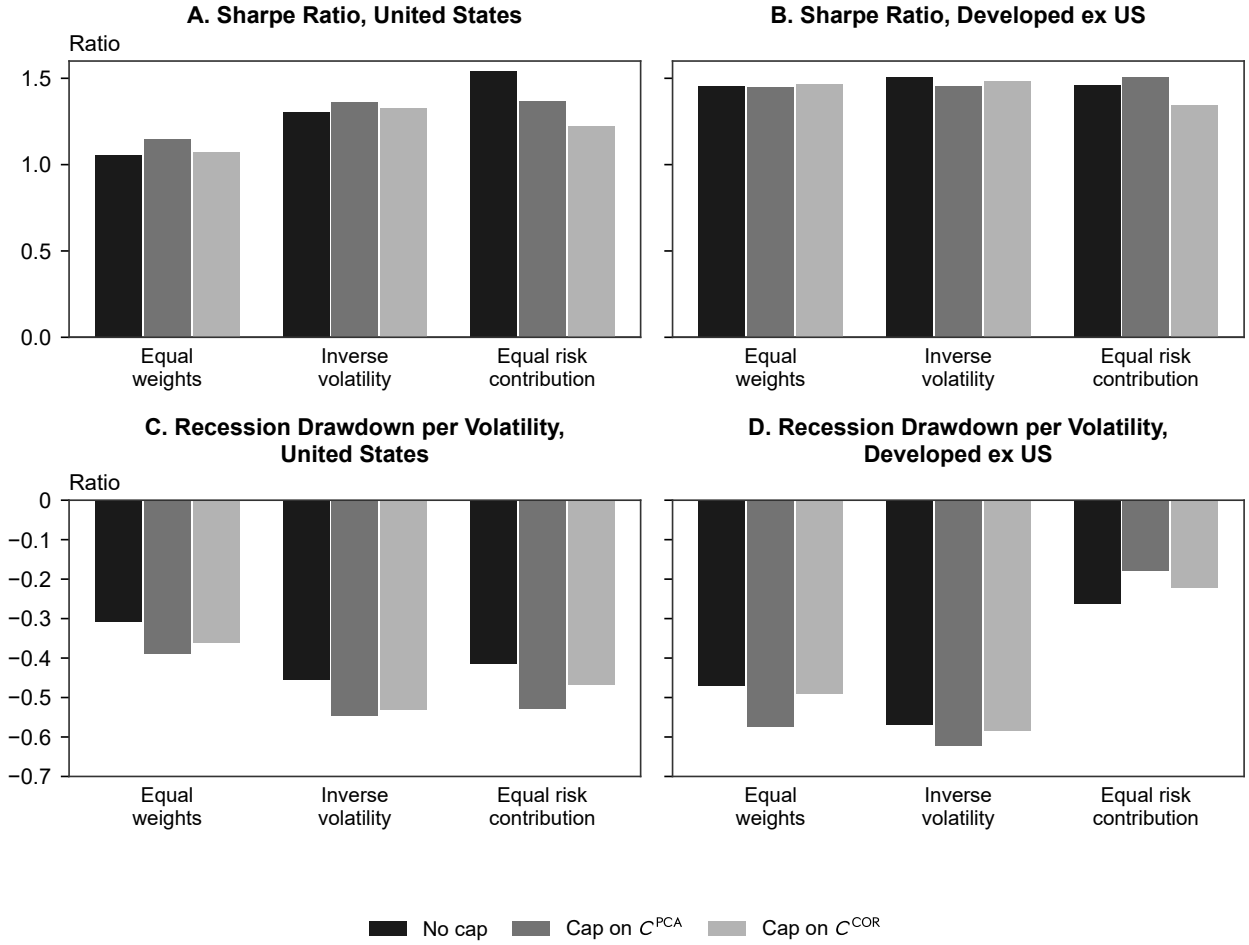
6.2 Sizing Factor Sleeves

Exhibit 7 compares the three sizing rules of Section 2.5, each with and without a crowding cap, by Sharpe ratio and by mean recession drawdown per unit of volatility; Exhibit A24 of the Appendix gives the tests.

For an equal-weighted investor, capping the most crowded factors helps in the United States, raising the Sharpe ratio from 1.05 to 1.14, mainly because the cap removes weight from the most volatile factors. Inverse-volatility weights do more (1.30), and equal risk contribution more still (1.54). On top of inverse-volatility weights, the cap changes little (1.36). On top of equal risk contribution, both caps significantly lower the Sharpe ratio (to 1.37 and 1.23). Outside the United States, no rule beats equal weights significantly (1.45 for equal weights, 1.50 for inverse-volatility weights, and 1.46 for equal risk contribution). On the anomalies, a cap on within-factor comovement slightly lowers the Sharpe ratio of equal weights (Section F of the Appendix).

In the United States, the higher Sharpe ratios of risk-based weights come with deeper recession losses per unit of risk: inverse-volatility weights and equal risk contribution lost 0.45 and 0.41 times their annual volatility on average, against 0.31 for equal weights. In developed markets, equal risk contribution lost less (0.26 against 0.47), over only three recessions. Every crowding cap on equal weights or inverse-volatility weights deepened these losses, in both regions. Only the cap on standard centrality applied to equal risk contribution in developed markets reduced them (0.18). A US risk budget thus allocated less capital to volatile factors, but it did not protect better, per unit of risk, when the cycle turned. Caps

Exhibit 7. Risk Budgets and Crowding Caps



Notes: JKP factors, monthly rebalancing, gross of costs. Panels A and B: annualized Sharpe ratio; Panels C and D: mean maximum drawdown during NBER recessions divided by the portfolio's annualized volatility (closer to zero is better). Equal weights, inverse-volatility weights, and equal risk contribution (shrunk covariance matrix), each without a cap and with a cap that halves the weight of the 30% of factors with the highest C^{PCA} or C^{COR}. United States: 1957–2024, ten recessions; developed ex US: 1991–2024, three recessions. Tests: Exhibit A24.

raise one-way turnover modestly (equal weights from 10% to 16% a year), and equal risk contribution trades most (30% a year in the United States and 35% in developed markets). Its volatility is low (0.75% a year in the United States), so matching the volatility of equal weights would require leverage of about 3.5. Levered to the volatility of inverse-volatility weights, its advantage of about 0.46% a year would be lost at trading costs above about 0.7% of the value traded. With the singular sample covariance matrix instead of the shrunk one, equal risk contribution earns less (1.33 in the United States; Section G of the Appendix).

6.3 Using the Measures in Practice

The results assign each measure one use. Trailing volatility and risk-based weights are the tools for sizing factor sleeves, with two caveats: their Sharpe gain is a US result, gross of costs, and in US recessions they lost more per unit of volatility. Risk-based sizing also relies on current volatility, and prolonged periods of low volatility have historically preceded banking crises (Danielsson et al. 2018). A risk budget is therefore best paired with stress tests rather than relied on alone.

Standard centrality is best read as a second volatility report. A rank correlation with each sleeve's trailing volatility near one, which holds even across nine themes (0.94; Exhibit A11), signals that the crowding report restates the risk report. The score should not be used as a positioning measure, a loss warning, a return forecast, or a cap on risk-based weights. Correlation-based centrality, comovement, and run-up measures are descriptive. In these data, they point to smaller, not larger, later losses relative to risk.

The cycle beta is the measure to watch next to volatility. It needs no principal components. It is the absolute slope of a regression of a sleeve's monthly returns on the innovation of a business-cycle indicator that combines valuation, credit, activity, and inflation information, estimated over at least three years and lagged three months to respect publication lags. Among US factors, sleeves with a high cycle beta lost more over the following year than their volatility implied, and their volatility tended to rise. It is a risk flag, not a recession forecast, best used to question a risk budget built on trailing volatility alone. The evidence comes from broad universes of 104 to 148 sleeves and holds firmly only for US factors; for books with a handful of sleeves, and for rules outside the United States, it is untested. Excess centrality shows where cyclical comovement sits among the sleeves, but it does not replace the cycle beta. Neither the cycle measures nor the regimes support return forecasts or rotation rules (Section 7).

7 Robustness and Scope of the Evidence

7.1 The Loss Signal across Decades and Definitions

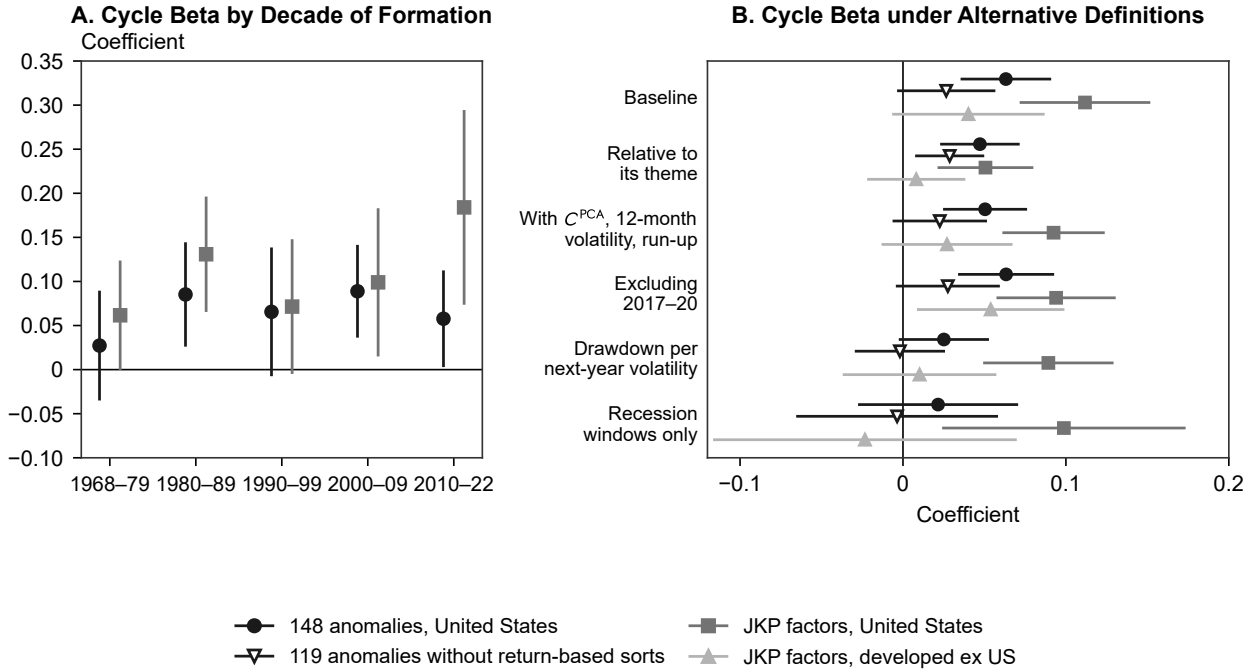
The loss signal of the cycle beta does not rest on one decade, one theme, or one set of controls (Exhibit 8). Among US factors, its coefficient is positive in each of the five decades since 1968, from 0.062 in 1968–79 to 0.184 in 2010–22, and significant in four of them; on the anomalies, it is positive in every decade and significant in three. It keeps its sign relative to the theme mean, with added controls, and without the formation months whose outcomes overlap the late part of value's long drawdown in 2018–20. Its rise over the decades may partly reflect estimation: the cycle beta comes from an expanding history, about eleven years long in 1968 and more than sixty by 2022. Panel B also shows where the signal is weak: in developed markets, among the anomalies without return-based sorts, and, outside the US factors, for the drawdown per unit of realized volatility and in windows that contain a recession. For excess centrality, the decade pattern is similar, and with the FRED-MD target its loss coefficients are positive in every sample but smaller in the United States (Exhibit A18 of the Appendix).

7.2 Factor Universes

Exhibit A13 of the Appendix repeats the main tests in two wider universes: with the 18 factors sorted on statistics of past returns (122 factors), and with all 153 JKP factors.

The volatility content of standard centrality does not depend on the universe (0.94 to 0.96 in both regions). In every universe, the recession drawdowns of crowded factors match those of volatile factors. Excess centrality carries more information about cycle sensitivity in the wider universes. Its US t-statistic is 3.3 with 122 factors and 3.8 with 153, against 1.9 in the main universe; pooled, it is 3.1 and 3.0, against 2.6; and beyond the cycle beta,

Exhibit 8. The Cycle-Beta Loss Signal across Decades and Definitions



Notes: Incremental rank-regression coefficient of the cycle beta $|\lambda|$ for the later loss relative to volatility, with the controls of Exhibit 3; 95% intervals from Newey–West standard errors (12 lags). Panel A: by decade of formation. Panel B: baseline; relative to the theme mean; with standard centrality, 12-month volatility, and the 12-month run-up as added controls; without formation months 2017–2020; outcome scaled by the volatility of the next 12 months instead of trailing volatility; and only formation months whose 12-month outcome window contains an NBER recession. Anomalies without return-based sorts: the 119 anomalies of Section 7. Checks added after the main results (Exhibit A14).

2.6 and 2.7, against 1.7. The reason is that the factors sorted on market and volatility risk carry much of the cyclical comovement that the target recovers. The main conclusions hold in all three universes. How strong excess centrality looks relative to the cycle beta depends on whether these factors are included.

The anomaly sample raises the reverse question, because it keeps 29 anomalies sorted on return seasonality, other statistics of past returns, or the price level, which the JKP universe excludes. In a check added after the main results, we rebuilt all measures on the remaining 119 anomalies. Standard centrality remains a volatility ranking (0.88), no crowding measure points to larger losses, and mean correlation and within-factor comovement still point to smaller ones (t of -4.5 and -3.6). As Section 5 reports, the cycle beta itself weakens, but the cycle share and excess centrality do not. The JKP factors also exclude such sorts

and keep a strong signal. The gap between the two US data sets therefore points to their construction: capped value-weighted terciles of larger stocks in the JKP data, mostly equal-weighted decile portfolios in the anomaly data. The cycle beta also carries more volatility content than the cycle share on these anomalies (0.48 against −0.13), and a version scaled by volatility keeps a significant signal (0.030, t of 2.3; Section F of the Appendix).

7.3 Windows, Inference, and Controls

The volatility result is the same for every estimation window from 36 to 120 months: 0.96 in the United States and 0.94 to 0.95 in developed markets (Exhibit A15). The information in excess centrality about cycle sensitivity is less stable. Its pooled t-statistics range from 1.8 to 2.6 across windows, and from 1.0 to 2.7 beyond the cycle beta. With 48 or 60 Newey–West lags instead of 24, they fall to 2.1 and 1.4 (Exhibit A16), and with more flexible controls, its t-statistics beyond the cycle beta are 1.3 to 1.7 (Exhibit A17). The loss signal is less sensitive to these choices. In checks added after the main results, with 24 or 36 lags instead of 12, the t-statistics of excess centrality for the later loss stay between 3.5 and 4.9 in all three samples. Scores standardized against their own trailing 36 months, as Kinlaw et al. (2019) use them, also track volatility. With the FRED-MD target, excess centrality also carries information about future cycle sensitivity, although its ranking is only weakly related to that of the main design (Exhibit A18). These checks confirm the volatility content and the loss results. They also show that the evidence for excess centrality beyond the cycle beta is fragile.

7.4 Other Markets, Data Sets, and Regimes

In emerging markets, standard centrality is again a volatility ranking (0.88), the crowding and volatility sorts give almost the same recession drawdown gap, and excess centrality adds no information about future cycle exposure. There, equal risk contribution with the sample covariance matrix earns less than equal weights, and less still with a crowding cap

(Exhibit A11). Across the nine US themes, the rank correlation with volatility is 0.94, and the two sorts again give almost the same gap. In the development sample of anomaly themes, standard centrality is significant beyond the controls for cycle sensitivity, driven by 1990–2024, whereas on the JKP factors it is not. Its volatility content (0.92) and its recession record are the same in both data sets (Supplemental Material).

The regimes of the indicator do not support rotation rules. Risk-adjusted performance differs across the four regimes for 27 of the 104 US factors, a borderline result (placebo p of 0.05) for an analysis specified after the main results (Exhibit A5). At the theme level, however, we cannot reject that performance is the same in every phase after a correction for multiple testing, and no theme has the same best and worst regime in developed markets (Exhibit A6). Diversification across themes weakens in downturns, but from a low level: the average pairwise correlation of the factors is 0.07 in expansions and recessions and 0.10 at the peak (Exhibit A7).

7.5 Scope of the Evidence

The evidence has clear limits. The scores are returns-based and say nothing directly about who holds a factor. Centrality is tested in levels across factor sleeves, not as the within-factor shifts that Kinlaw et al. (2019) use, and vendor models are not tested. Within-factor comovement is measured with three decile portfolios per leg rather than individual stocks, on anomalies that exclude momentum, where comovement-based crowding is best documented. "Normal risk" is a slow 60-month volatility, so the loss outcome also reflects changes in volatility. Outside the US factors, the signal of the cycle beta is mainly such a change. The evidence comes from universes of 104 to 148 sleeves; whether it carries over to books of a few sleeves is not tested. The indicator is a US indicator applied to all regions, uses current data vintages, and has few recovery months. The main universe was chosen after first results, and the comparison of crowding measures for losses was added last. The cycle beta is itself

estimated, and errors in it and in the outcomes can blur the comparison between measures. Finally, the factors are published anomalies, whose premia tend to be lower after publication (McLean and Pontiff 2016), and long–short portfolios before trading, shorting, and financing costs. Whether the results carry over to long-only factor sleeves is not tested.

8 Conclusion

Crowding reports promise to tell investors which factors are dangerous. We find that, built from returns, they mostly tell investors which factors are volatile.

Standard centrality ranks factors as volatility does, with rank correlations of 0.96 in the United States and 0.94 in developed markets on 104 factors, and 0.88 on 148 anomalies. The factors it labels crowded lose in recessions what volatile factors lose, and they do not move together. Comovement across and within factors and past run-ups are nearly unrelated to volatility. But they point to factors that later lose less, not more, relative to their risk.

The information about later losses lies in what a factor's common risk is tied to, not in how large it is. Among US factors, a simple cycle beta flags the factors that later lose more than their volatility implies: the top 30% by cycle beta lost 3.2 percentage points more over the next year than the bottom 30%, or 0.08 more per unit of annual volatility. The signal holds with one observation a year and after controls for centrality, short-term volatility, and run-up. Outside the US factors, the evidence is weaker. The US magnitude is rejected in developed markets, and among the anomalies the signal is clearest in a share-of-risk version of the cycle beta and comes mostly from rising volatility. Directing the crowding decomposition at the cycle with target principal component analysis reproduces the US signal but adds nothing beyond the cycle beta. No crowding measure is a profitable return signal, crowding caps add little once weights account for risk, and the regimes do not support rotation rules.

For investors, three practical rules follow. Read a crowding report next to the risk report, and discount it when the two rank sleeves alike. Size sleeves in the risk budget rather than with crowding caps. And watch each sleeve's cycle beta next to its volatility, as a flag, among US factors, for losses that volatility alone does not show. Stock-level comovement, regional business-cycle indicators, and holdings or flow data are the natural next tests. A crowding report that ranks sleeves as the risk report does is a second risk report, not a warning.

Declaration of Interest Statement

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Declaration of generative AI use

The author used a generative AI assistant (Claude, Anthropic) for programming support in the data-processing and analysis code, for literature searches, and for language editing of the manuscript. The research questions, the design and pre-registration of the tests, and the interpretation of the results are the author's; all code, results, citations, and text were checked by the author, who takes full responsibility for the content of this article.

Data availability

Factor returns are publicly available from Jensen, Kelly, and Pedersen (https://jkpfactors.com); anomaly returns and daily decile portfolio returns of the development sample from the Open Source Asset Pricing project (https://www.openassetpricing.com); the monthly and daily Fama–French factors from Kenneth French's data library (https://mba.tuck.dartmouth.edu/pages/faculty/ken.french/data_library.html; daily file built from the September 2026 CRSP database); the inputs to the business-cycle indicator from Robert Shiller's online data (http://www.econ.yale.edu/~shiller/data.htm) and the Federal Reserve Bank of St. Louis (FRED, https://fred.stlouisfed.org); and the recession dates from the NBER (https://www.nber.org/research/data/us-business-cycle-expansions-and-contractions). The indicator uses the FRED series AAA, BAA, and INDPRO and Shiller's price, earnings, and consumer-price series. The robustness target of the Appendix uses the FRED-MD database of the Federal Reserve Bank of St. Louis (https://www.stlouisfed.org/research/economists/mccracken/fred-databases; vintage of December 2025). The analysis was carried out in Python (pandas, NumPy, and statsmodels); the code is not publicly available.

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